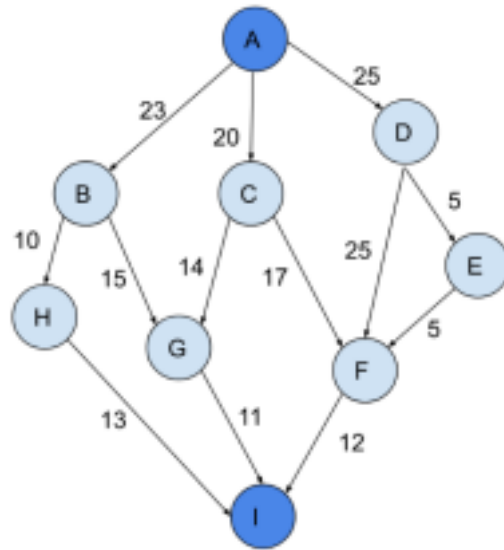


Graded Quiz 1(Set A)



Question 1 [Marks 10]

- For each of the nodes from A to I in the graph above, define a random heuristic value that is consistent. [7.5]
- Considering A as the start node and I as the goal node, Apply A* search algorithm for the given graph. Make sure to show each step. [7.5]

Solution:

a.

Heuristic	Neighbour 1	Neighbour 2	Neighbour 3	Neighbour 4	Neighbour 5
$h(A)=20$	$\text{Cost}(A,B)+h(B)$ $=23+10=33$	$\text{Cost}(A,C)+h(C)$ $=20+14=34$	$\text{Cost}(A,D)+h(D)$ $=25+5=30$	N/A	N/A
$h(B)=10$	$\text{Cost}(B,H)+h(H)$ $=10+12=22$	$\text{Cost}(B,G)+h(G)$ $=15+10=25$	N/A	N/A	N/A
$h(C)=14$	$\text{Cost}(G,C)+h(G)$ $=14+10=24$	$\text{Cost}(F,C)+h(F)$ $=17+11=28$	N/A	N/A	N/A
$h(D)=5$	$\text{Cost}(E,D)+h(E)$ $=5+5=10$	$\text{Cost}(F,D)+h(F)$ $=25+11=36$	N/A	N/A	N/A
$h(E)=5$	$\text{Cost}(E,F)+h(F)$ $=5+11=16$	N/A	N/A	N/A	N/A
$h(F)=11$	$\text{Cost}(I,F)+h(I)$ $=12+0=12$	N/A	N/A	N/A	N/A
$h(G)=10$	$\text{Cost}(I,G)+h(I)$ $=11+0=11$	N/A	N/A	N/A	N/A
$h(H)=12$	$\text{Cost}(I,H)+h(I)$ $=13+0=13$	N/A	N/A	N/A	N/A

$h(1)=0$	N/A	N/A	N/A	N/A	N/A
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b. Fringe: $[A^{20}]$

NODE: A^{20}

Fringe: $[B^{10}, C^{14}, D^5]$

NODE: A^{20}

Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

Fringe: $[B^{10}, C^{14}, F^{11},]$

NODE: E^5

Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

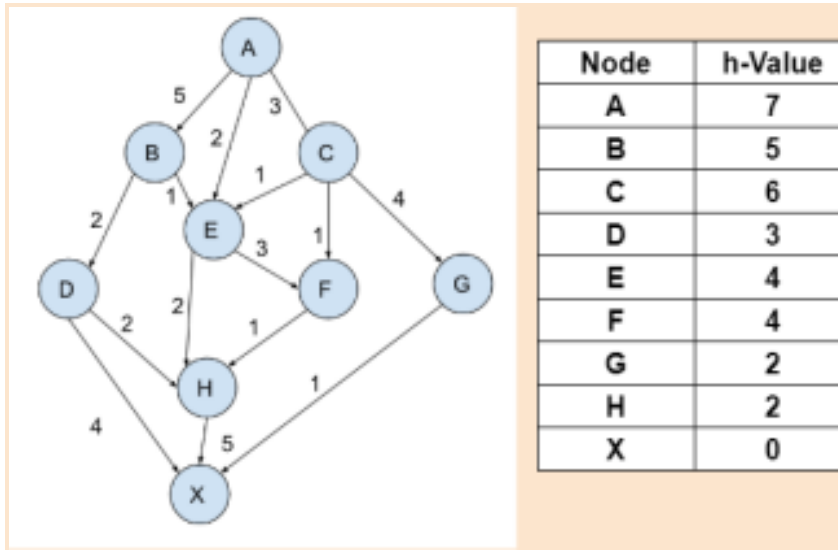
Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

Fringe: $[B^{10}, C^{14}, E^5, F^{11}]$

NODE: D^5

Graded Quiz 1(Set B)



Question 1 [Marks 10]

In the above graph, node A is the source, and node X is the destination. The arrows indicate directed edges. The table contains the heuristic values of each node. Now answer the following:

- Apply** A* algorithm to find the path from the source to the destination. Show the steps. In case, you end up with multiple nodes with $f(n) = g(n) + h(n)$, then you can break the tie by choosing the lexicographically (alphabetically) smaller node. Suppose, node C and node D has the same $f(n)$, in that case, choose C. [7.5]
- Is the heuristic consistent and admissible? Why or why not? **Explain** with appropriate calculations. [7.5]

SOLUTION

Question 1

A. Fringe: $[A^7]$

NODE: A^7

Fringe: $[B^{10}, C^9, E^6]$

NODE: A^7

Fringe: $[B^{10}, C^9, F^9, H^6]$

NODE: E^6

Fringe: $[B^{10}, C^9, F^9, X^9]$

NODE: H^6

Fringe: $[B^{10}, F^9, X^9, E^8, F^8, G^9]$

NODE: C^9

Fringe: $[B^{10}, F^9, X^9, F^8, G^9, F^{11}, H^8]$

NODE: E^8

Fringe: $[B^{10}, F^9, X^9, G^9, F^{11}, H^8, H^{10}]$

NODE: F^8

Fringe: $[B^{10}, F^9, X^9, G^9, F^{11}, H^{10}, X^{13}]$

NODE: H^8

Fringe: $[B^{10}, X^9, G^9, F^{11}, H^{10}, X^{13}, H^8]$

NODE: F^9

Fringe: $[B^{10}, X^9, G^9, F^{11}, H^{10}, X^{13}, X^{13}]$

NODE: H^8

Fringe: $[B^{10}, X^9, F^{11}, H^{10}, X^{13}, X^{13}, X^8]$

NODE: G^9

Fringe: $[B^{10}, X^9, F^{11}, H^{10}, X^{13}, X^{13}]$

NODE: X^8

Optimal PATH: A->C->G->X

B.

Heuristic	Neighbour 1	Neighbour 2	Neighbour 3	Neighbour 4	Neighbour 5	Consistent?	Lowest Cost	Admissible?
$h(A)=7$	$\text{Cost}(A,B)+h(B)$ $=5+5=10$	$\text{Cost}(A,E)+h(E)$ $=2+4=6$	$\text{Cost}(A,C)+h(C)$ $=3+6=9$	N/A	N/A	No	8	Yes
$h(B)=5$	$\text{Cost}(B,D)+h(D)$ $=2+3=5$	$\text{Cost}(B,E)+h(E)$ $=1+3=4$	N/A	N/A	N/A	No	6	Yes
$h(C)=6$	$\text{Cost}(A,C)+h(A)$ $=3+7=10$	$\text{Cost}(E,C)+h(E)$ $=1+4=5$	$\text{Cost}(F,C)+h(F)$ $=1+4=5$	$\text{Cost}(G,C)+h(G)$ $=4+2=6$	N/A	No	5	No
$h(D)=3$	$\text{Cost}(H,D)+h(H)$ $=2+2=4$	$\text{Cost}(X,D)+h(X)$ $=4+0=4$	N/A	N/A	N/A	Yes	4	Yes
$h(E)=4$	$\text{Cost}(H,D)+$	$\text{Cost}(H,D)+$	N/A	N/A	N/A	Yes	7	Yes

	$h(H)$ $=2+2=4$	$h(H)$ $=2+2=4$						
$h(F)=4$	$\text{Cost}(H,F)+h(H)=1+2=3$	N/A	N/A	N/A	N/A	No	6	Yes
$h(G)=2$	$\text{Cost}(X,G)+h(X)=1+0=5$	N/A	N/A	N/A	N/A	No	1	No
$h(H)=2$	$\text{Cost}(X,H)+h(X)=5+0=5$	N/A	N/A	N/A	N/A	Yes	5	Yes
$h(X)=0$	N/A	N/A	N/A	N/A	N/A	Yes	0	Yes